

REVIEW

Recent Developments in Polysulfone Studies

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ABSTRACT

This review provides a comprehensive overview of the synthesis methods for different types of polysulfones, including bisphenol A-based polysulfone (PSF), polyethersulfone (PES), and polyphenylsulfone (PPSU). Conventional polycondensation techniques, including solution-phase and melt-phase processes, are critically analyzed alongside innovative synthetic approaches such as vortex fluidic-mediated polymerization, phase-transfer catalyzed polycondensation, and AB-type monomer self-polycondensation, which collectively enable precise control over macromolecular architectures while enhancing production efficiency. Furthermore, this review highlights significant advancements in sustainable polymer chemistry, particularly the development of bio-based synthetic routes utilizing renewable monomer feedstocks to mitigate dependence on petrochemical precursors. Special emphasis is placed on advanced functionalization strategies, including surface modification and copolymerization techniques, which impart tailored properties such as enhanced hydrophilicity, tunable ion-exchange capacity, and optimized biocompatibility—critical parameters for next-generation filtration membranes and implantable medical devices. These scientific and technological innovations underscore the pivotal role of green chemistry principles and novel polymerization paradigms in advancing the field of high-performance polysulfones, addressing the escalating demand for environmentally benign materials with superior engineering characteristics.

1 | Introduction

Polysulfones (PSUs) represent an essential class of high-performance engineering thermoplastics, with exceptional mechanical strength, thermal stability, and chemical resistance. These attributes arise from the unique molecular structure of polysulfones, which comprises aromatic groups linked by sulfone ($-\text{SO}_2-$) and ether ($-\text{O}-$) bonds. This combination bestows polysulfones with remarkable resistance to high temperatures, oxidation, and aggressive chemical environments, enabling their widespread use in demanding fields such as aerospace, electronics, medical devices, and membrane separation technology [1, 2].

Polysulfones are well known for their ability to maintain mechanical and dimensional stability even in harsh environments, where temperatures can exceed 190°C . In some cases, the glass transition temperature (T_g) of certain polysulfones reaches as high as 220°C , and the polymers find high-temperature applications [3]. Their low water absorption, combined with good strength and chemical resistance, has made them indispensable in industries. Additionally, their excellent dielectric properties and flame retardancy make polysulfones ideal candidates for use as insulation materials in electrical and electronic applications.

Based on monomer compositions polysulfones can be categorized into several main types: bisphenol A-based polysulfone

(PSF), polyether sulfone (PES), and polyphenylsulfone (PPSU). Each of these types exhibits distinct molecular structures that impart specific physical and chemical properties, allowing them to meet varying industrial requirements [2]. For instance, bisphenol A-based polysulfones offer exceptional heat and mechanical resistance, making them suitable for electronics and medical devices, while polyphenylsulfones are known for their higher impact resistance and chemical durability, which are critical in more aggressive environments [1].

In recent years, polysulfones have attracted significant attention in membrane technology, particularly in water treatment, gas separation, and fuel cell applications. Their high chemical resistance and selectivity have made polysulfone-based membranes a preferred choice for advanced filtration and separation systems [4]. Polysulfones are important thermoplastics used for the fabrication of polymer membranes. Despite significant advancements in synthetic methodologies and functional applications, polysulfones exhibit several material limitations that warrant consideration. These include susceptibility to solvent-induced stress cracking due to their amorphous nature, and compromised weathering performance resulting from photo-oxidative degradation of aromatic moieties upon prolonged UV exposure. These intrinsic material constraints necessitate further molecular engineering to enhance environmental stability while maintaining the polymers' exceptional thermomechanical properties. Hence, the functionalization of polysulfones, achieved through post-polymerization modification or the incorporation of specific functional groups during synthesis, has further broadened their applications by enhancing properties such as hydrophilicity, conductivity, and fouling resistance.

Because of the growing industrial demand for high-performance materials, the synthesis and processing of polysulfones have become crucial for optimizing their properties. Different types of polysulfones require distinct synthesis approaches, primarily through nucleophilic aromatic substitution reactions or electrophilic aromatic substitution reactions [5]. These synthesis methods are directly related to the molecular weight, structural configuration and the performance characteristics of the resulting polymers. Therefore, understanding the synthesis methods for each type of polysulfone is critical for improving their functional capabilities and extending their industrial applications.

This review aims to provide a comprehensive overview of the various synthesis methods for different types of polysulfones, focusing on the key features of each approach and their impact on the physical and chemical properties of the final products. By examining these methods, we hope to offer valuable insights into the optimization of polysulfone production for advanced industrial applications.

To provide a clear overview and facilitate comparison, the main synthesis methods for polysulfones discussed in this review, along with their characteristic reaction conditions, advantages, and limitations, are summarized in Table 1.

The following sections will delve into the specifics of these methods for each type of polysulfone.

2 | Bisphenol A-Based Polysulfone (PSF)

Bisphenol A-based polysulfone (PSF) is synthesized through the polycondensation reaction of bisphenol A and dichlorodiphenyl sulfone under basic conditions. The molecular structure consists of aromatic rings connected by ether bonds, with these rings further linked through sulfone groups ($-\text{SO}_2-$). This configuration endows PSF with exceptional chemical and thermal stability. The presence of bisphenol A in the backbone provides rigidity to the polymer chain, while the sulfone groups enhance its oxidative and thermal resistance, enabling it to withstand extreme environmental conditions.

The fundamental structure of PSF is shown in Figure 1. Due to the bulky size of the sulfone groups, crystallization of the polymer chains is hindered, resulting in an amorphous morphology. This lack of crystallinity is a key characteristic of PSF, which determines the dimensional stability and resistance to stress cracking in various industrial applications [6].

Bisphenol A-based polysulfone (PSF) is a typical high-performance plastic due to its exceptional thermal stability, chemical resistance, and mechanical strength. With a glass transition temperature (T_g) ranging from 190°C to 220°C, PSF maintains structural integrity under high-temperature conditions, making it an ideal candidate for applications in extreme environments [7]. In addition to the chemical resistance, PSF exhibits outstanding mechanical properties, including high tensile strength and rigidity, and excellent impact resistance. This combination of properties is critical in cases where durability is paramount, such as in the automotive, aerospace, and industrial equipment industries [3, 8, 9]. PSF is also a transparent material with excellent optical properties, making it highly applicable in fields where visibility is essential, such as in medical devices and food processing equipment [10]. The combination of its mechanical, chemical, and optical characteristics makes PSF a versatile material for a broad range of applications.

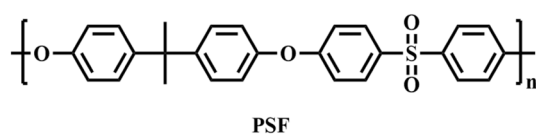
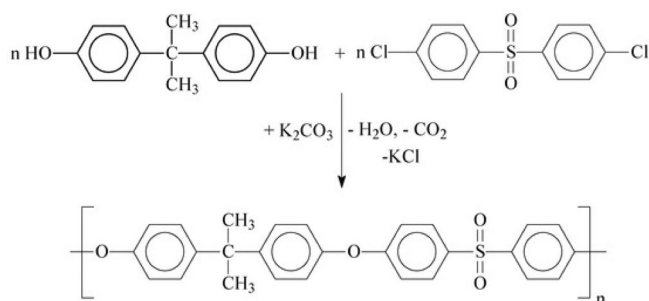
As the demands for performance and sustainability, the methods of PSF synthesis have been upgraded to improve functional properties, environmental friendliness, and production efficiency. This review highlights several methods used for PSF preparation, including traditional solution polycondensation, melt polycondensation, green synthesis approaches, and functional modifications of the polymers.

2.1 | Traditional Solution Polycondensation

The most prevalent industrial-scale production of PSF employs a nucleophilic aromatic substitution ($\text{S}_{\text{N}}\text{Ar}$) polyetherification process. Under strictly anhydrous conditions, the polycondensation reaction proceeds between bisphenol A (BPA) and 4,4'-dichlorodiphenyl sulfone (DCDPS) in aprotic polar solvents such as dimethylacetamide (DMAc). This step-growth polymerization involves the deprotonation of bisphenolic hydroxyl groups by alkaline bases (typically potassium carbonate), generating phenolate anions that subsequently undergo nucleophilic displacement of the activated aromatic chlorides on DCDPS, with concomitant formation of ether

TABLE 1 | Comparison of predominant synthetic methods for polysulfones.

Synthetic method	Reaction conditions	Advantages	Disadvantages	Typical polymers
Solution polycondensation	Aprotic solvent (DMAc, NMP), base (K_2CO_3), anhydrous	High M_w , narrow PDI, precise control, well-established	Large solvent volumes, salt byproduct, energy-intensive recovery, high cost	PSF, PES, PPSU
Melt/bulk polycondensation	Solvent-free, high temperature, catalyst-free options	No solvent, scalable, eco-friendly, low cost	Requires high-purity monomers, challenging viscosity control	PSF
Vortex fluidic device (VFD)	VFD reactor, controlled shear, temperature, anhydrous	Solvent-free, rapid, high M_w , enhanced T_g	Specialized equipment, not yet scaled up	PSF
Phase transfer catalysis (PTC)	Biphasic system, PTC catalyst, mild conditions	Avoids strict stoichiometry, ambient conditions	Catalyst residue possible	PSU
AB-type self-polycondensation	AB monomer, high temperature, no catalyst	Perfect stoichiometry, no byproducts, low PDI, green	Complex monomer synthesis	PES
Electrophilic substitution	Lewis acid catalyst ($AlCl_3$), Friedel-Crafts reaction	Alternative route	Branched structures, difficult to control linearity	PES
Nickel-catalyzed coupling	Ni(0) catalyst, ligand (PPh_3), aprotic solvent, inert atm.	High M_w , excellent thermal/oxidative stability	Requires specialized catalysts and conditions	PPSU

**FIGURE 1** | Structure of PSF.**FIGURE 2** | A polycondensation reaction between bisphenol A (BPA) and 4,4'-dichlorodiphenyl sulfone (DCDPS) [11].

linkages and evolution of potassium chloride as a byproduct (Figure 2) [11]. The reaction mechanism proceeds via a Meisenheimer complex intermediate, facilitated by the strong electron-withdrawing nature of the sulfone group, which activates the aryl chloride moieties toward nucleophilic attack. The solvent system must maintain rigorous anhydrous

conditions to prevent hydrolysis of both the monomers and the reactive intermediates while ensuring sufficient solubility of the growing polymer chains. This well-established synthetic route yields high-molecular-weight polysulfones with controlled molecular architectures and desirable thermomechanical properties.

In addition to conventional BPA monomers, cyclic structural analogs such as bisphenol Z (BPZ) have been successfully incorporated into PSF synthesis (Figure 3) [11]. These modified bisphenol derivatives, when copolymerized or blended with acrylic resin systems, exhibit significantly improved material characteristics including enhanced thermo-mechanical stability and processability. Through systematic optimization of solution polycondensation parameters—including monomer stoichiometry, reaction temperature, and catalyst selection—advanced PSF resin formulations with exceptionally low dielectric constants ($D_k < 2.1$) and minimal dielectric loss ($D_f < 0.007$) have been achieved. These engineered dielectric properties position these modified PSF materials as promising candidates for next-generation high-frequency electronic applications, particularly in 5G communication systems and high-speed printed circuit boards.

The synthesis versatility highlighted the possibility of PSF in a wide range of applications [12]. By fine-tuning the reaction parameters, the properties of the polymer can be controlled to meet

specific performance requirements for advanced technological applications. Furthermore, the kinetics and mechanisms of the polycondensation reactions, particularly between aryl halides

and BPA to form poly(arylene ether sulfone)s (PAES), have been extensively studied. Comparisons between aryl chloride (ArCl) and aryl fluoride (ArF) reactions reveal that ArF-based reactions follow a third-order rate law, whereas ArCl-based reactions follow a second-order rate law. Both experimental and computational studies provide valuable insights into the role of metal cations in facilitating these polycondensation reactions.

Moreover, a novel approach for recycling polycarbonates (PCs) into poly(sulfone)s (PSUs) through a one-step depolymerization and polycondensation process has gained attention (Figure 4) [11]. In this method, carbonate salt was employed to depolymerize PCs, after which bis(aryl fluorides) reacted with the resulting phenoxide intermediates to form high-performance PSUs. This process was particularly significant for its potential in repurposing polycarbonate waste, such as CDs or baby bottles, into valuable engineering thermoplastics, without necessitating intermediate purification steps. Computational studies have further elucidated the role of carbonate salts in facilitating both depolymerization and polycondensation.

Despite the emergence of novel techniques, traditional solution polycondensation remains the most widely adopted industrial method for PSF production due to its well-established

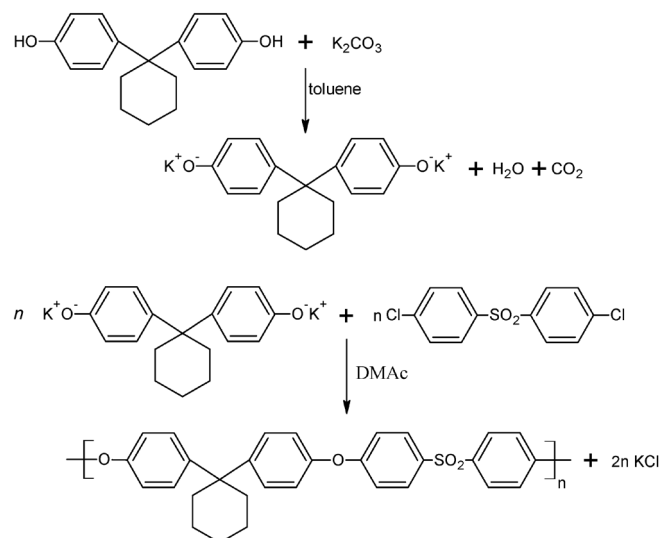


FIGURE 3 | Reaction scheme for the synthesis of bisphenol Z poly-sulfone (PSU-BPZ) [11].

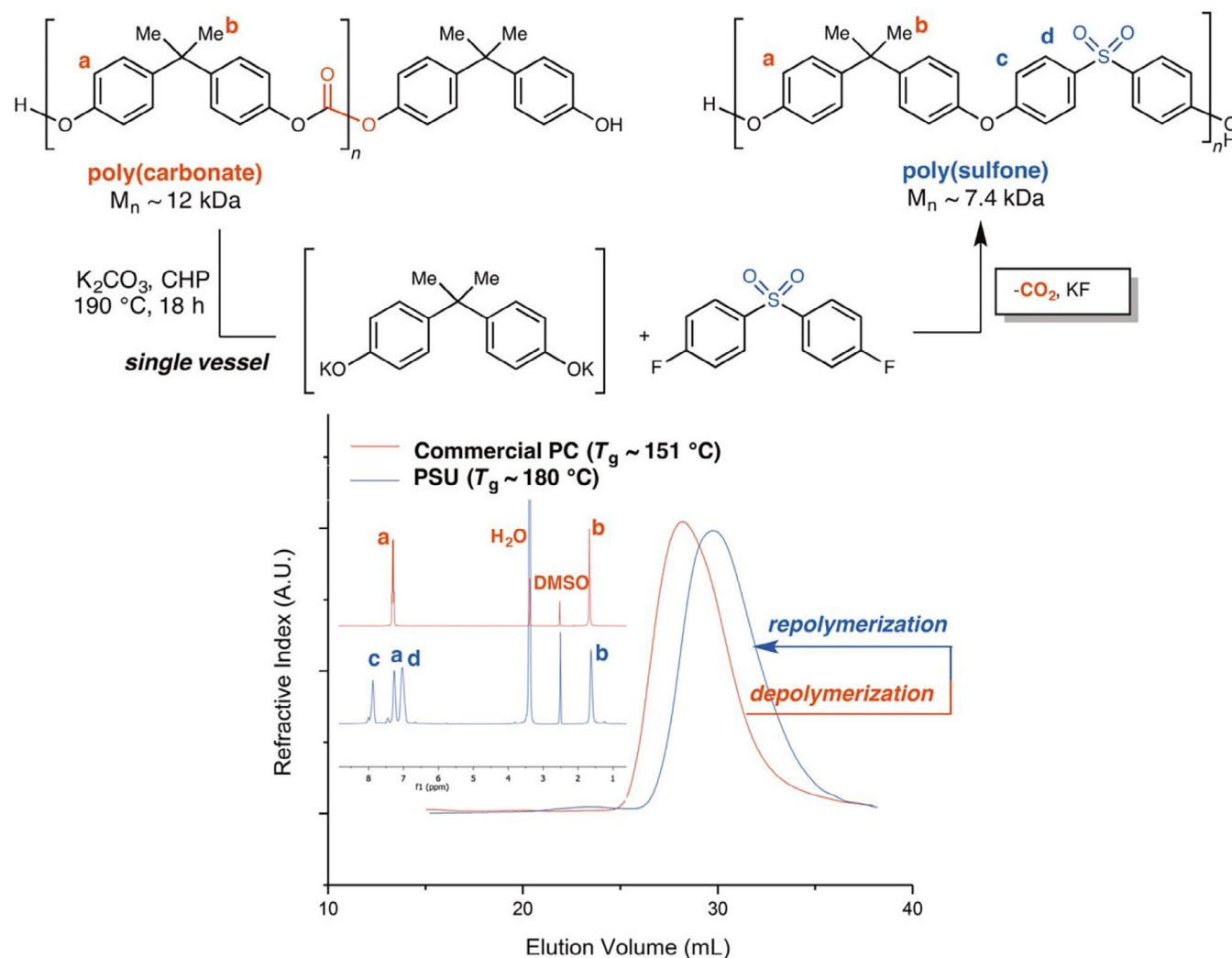


FIGURE 4 | Recycling of polycarbonate (PC) to poly(sulfone) (PSU) through a one-step depolymerization and polycondensation process [11].

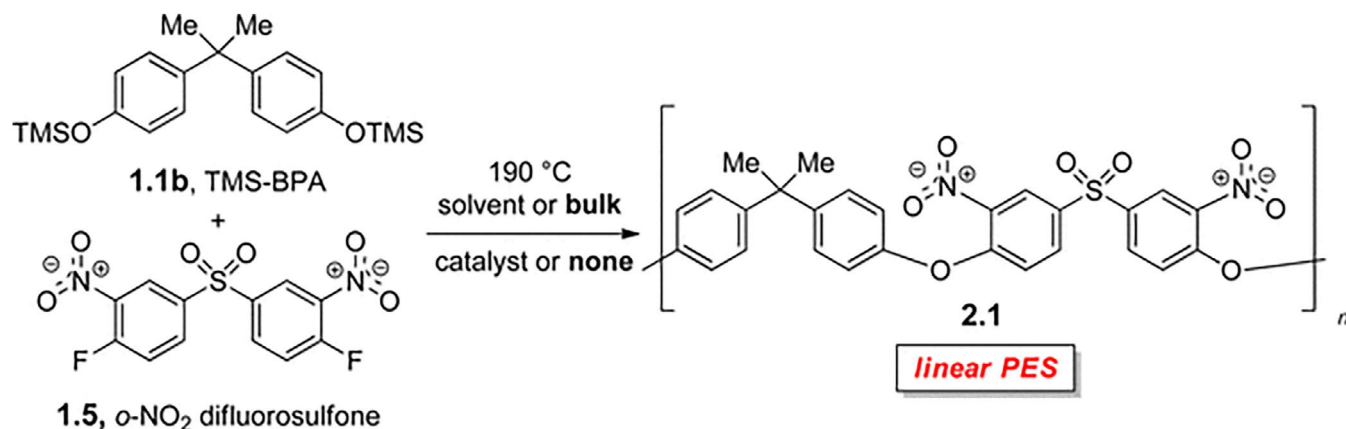


FIGURE 5 | Optimization of conditions for linear poly(aryl ether sulfone) synthesis [13].

protocols, excellent reproducibility, and capability to produce high-molecular-weight polymers with narrow polydispersity. The main advantages of this method include the precise control of the reaction kinetics and molecular structure by adjusting the solvent polarity, alkali concentration and temperature. However, its industrial application is hampered by several drawbacks: (1) the requirement for large volumes of expensive, high-boiling-point aprotic solvents (e.g., DMAc, NMP), which necessitate energy-intensive recovery and purification processes; (2) the generation of inorganic salt byproducts (e.g., KCl) that must be removed from the polymer solution; and (3) the need for strictly anhydrous conditions to prevent hydrolysis of monomers and intermediates, increasing operational complexity and cost. Therefore, while solution condensation performs well in lab-scale synthesis and production of high-precision materials, its environmental and economic costs drive exploration of more sustainable alternatives to mass production.

2.2 | Melt/Bulk Polycondensation

Melt polycondensation is a highly efficient, solvent-free polymerization method extensively used in the industrial production of bisphenol A-based polysulfones (PSFs). This approach is favored due to its scalability and the elimination of solvent-related issues, making it ideal for large-scale manufacturing. Recently, a novel method for synthesizing linear polysulfone (PSF) under catalyst-free and solvent-free conditions was introduced, utilizing Meisenheimer complexes as the key reactive intermediates. The process employed nitro-substituted aryl fluorides and trimethylsilyl-protected bisphenol A (TMS-BPA) in a high-temperature polycondensation reaction, successfully producing linear PES (Figure 5) [13].

Bulk polymerization, classified as a solvent-free polymerization technique, is characterized by the direct thermal polymerization of monomers in their neat state to yield polymeric materials. This method demonstrates particular advantages for industrial-scale production due to the absence of solvent-mediated side reactions and the consequent elimination of solvent removal and purification steps. The inherent catalyst-free nature of this process, coupled with its minimal environmental footprint, renders it both economically favorable and

sustainable. Notably, bulk polymerization enables the efficient synthesis of high-performance polysulfones with controlled molecular weights and narrow polydispersity indices, while simultaneously suppressing undesirable byproduct formation through precise temperature modulation and reaction stoichiometry control.

2.3 | Vortex Fluidic Mediated Synthesis

The use of a vortex fluidic device (VFD) was demonstrated to be an efficient approach to the synthesis of polysulfone (PSF) [14]. By precisely controlling parameters such as rotational speed, tilt angle, and temperature, the VFD method is able to produce high-molecular-weight polysulfones while optimizing key material properties, including thermal stability and T_g (Figure 6).

The process, carried out under anhydrous conditions, leveraged the high shear forces generated within the VFD to produce PSF without any chlorinated solvents [15]. As compared to the conventional synthesis methods, the high-shear environment not only facilitated polymerization but also achieved higher molecular weights and improved T_g values. This approach presents a highly efficient, one-step process for PSF production, yielding materials with superior properties while minimizing environmental impact by avoiding harmful solvents.

2.4 | Phase Transfer Catalyzed Reaction

The phase transfer-catalyzed (PTC) copolymerization of polyisobutylene (PIB) and polysulfone (PSU) represents an innovative approach to Williamson polyether synthesis, wherein the catalytic system facilitates interfacial reaction kinetics between immiscible phases [16]. This methodology offers distinct advantages over conventional polycondensation techniques by obviating the requirement for precise stoichiometric balance through the dynamic equilibrium established by the phase transfer agent. The PTC-mediated mechanism enables efficient coupling of nucleophilically functionalized PIB macromonomers (typically hydroxyl- or alkoxide-terminated) with electrophilic PSU precursors under ambient conditions, while maintaining excellent control over molecular weight distribution.

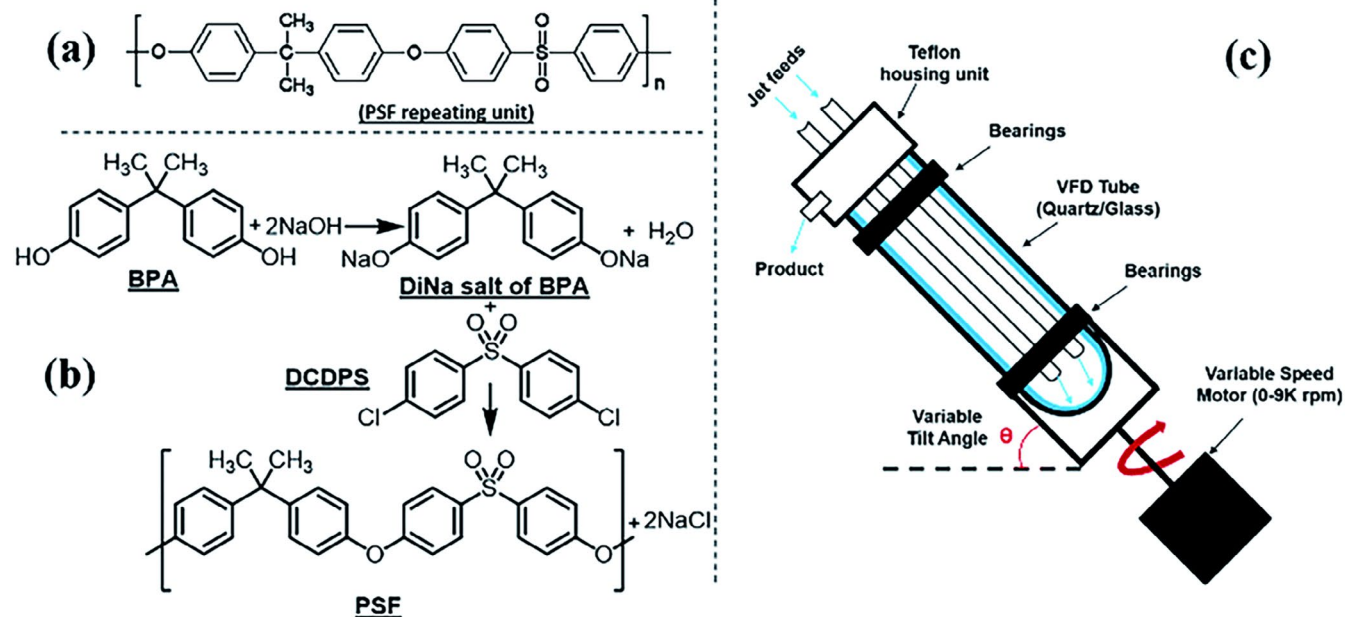


FIGURE 6 | (a) Aromatic PSF repeating unit in PSF, (b) Schematic of PSF synthesis involving nucleophilic substitution, and (c) Diagram of the vortex fluidic device (VFD) [14].

2.5 | Chemical Functionalization and Green Synthesis Strategies

Recently, researchers have been actively exploring green synthesis strategies to replace traditional petrochemical-based methods for the synthesis of BPA [17]. One notable advancement involved the use of lignin-based monomers as a renewable substitute for conventional BPA in the development of bio-based polysulfones. This biomass-derived approach not only reduced dependence on petrochemical resources but also mitigated the environmental and health risks associated with BPA. Wadgaonkar and co-workers demonstrated the applications of bio-based polymers in nucleophilic aromatic substitution (S_NAr) reactions (Figure 7) [18]. By modifying aromatic rings with natural material-derived structural units, this method maintained the excellent mechanical properties and thermal stability of polyethersulfone (PES) while enhancing its biodegradability and environmental friendliness. This innovation presents a promising pathway toward the synthesis of sustainable, high-performance polymers. They synthesized a fully bio-based bisphenol monomer, 4,4'-(furan-2-ylmethylene)bis(2-methoxyphenol) (BPF), through a base-catalyzed condensation of furfural and guaiacol. Both precursors are derived from lignocellulosic biomass, and the reported yield reached 58%. This BPF monomer was then copolymerized with BPA and bis(4-fluorophenyl) sulfone via S_NAr polycondensation, yielding a series of partially bio-based poly(ether sulfone)s.

Notably, these bio-based polymers attained high molecular weights ($\bar{M}_n$ up to 131 000 g mol⁻¹, η_{inh} ~1.47 dL g⁻¹). Their thermal and mechanical properties rivaled those of petroleum-based analogs, with glass transition temperatures (T_g) ranging from 179°C–190°C, tensile strength between 76.7–83.8 MPa, and Young's modulus of 1.02–1.18 GPa. This confirms that sustainable feedstocks can be incorporated without sacrificing the high-performance traits vital for engineering applications.

In addition to the green synthesis, chemical functionalization has become an essential strategy for enhancing the performance of BPA-based polysulfones, particularly in the applications of membrane technologies and biomedical devices [19]. Recent research focused on the functionalization of polysulfone membranes by introducing various functional groups during synthesis, improving their resistance to protein adsorption and enhancing biocompatibility [20]. These functional modifications have enabled polysulfone membranes to exhibit superior interactions with biomolecules, making them particularly suitable for medical applications such as hemodialysis. A noteworthy development in polysulfone technology was the creation of a novel multiblock copolymer designed for CO₂ capture using a moisture swing mechanism [21]. The copolymer consisted of polydiallylammonium (PDADMA) blocks for CO₂ capture and polysulfone (PSF) blocks for mechanical stability. This unique combination allowed the copolymer to efficiently absorb and release CO₂ under varying moisture conditions, presenting a viable solution for direct air capture (DAC) technologies.

In the realm of membrane enhancement, incorporating cellulose nanofibers into polysulfone matrices has significantly improved mechanical properties and water permeability [22]. This strategy demonstrates great potential for the functionalization of polysulfones through blending or adding nanomaterials, particularly in water treatment applications where both strength and permeability are critical.

In summary, traditional polycondensation remains a fundamental method for synthesizing bisphenol A-based polysulfones. However, emerging technologies such as vortex fluidic synthesis and chemical functionalization strategies are paving new paths for improving material properties and production efficiency. Melt polycondensation, in particular, is an efficient and environmentally friendly industrial method, offering distinct advantages in the large-scale synthesis of

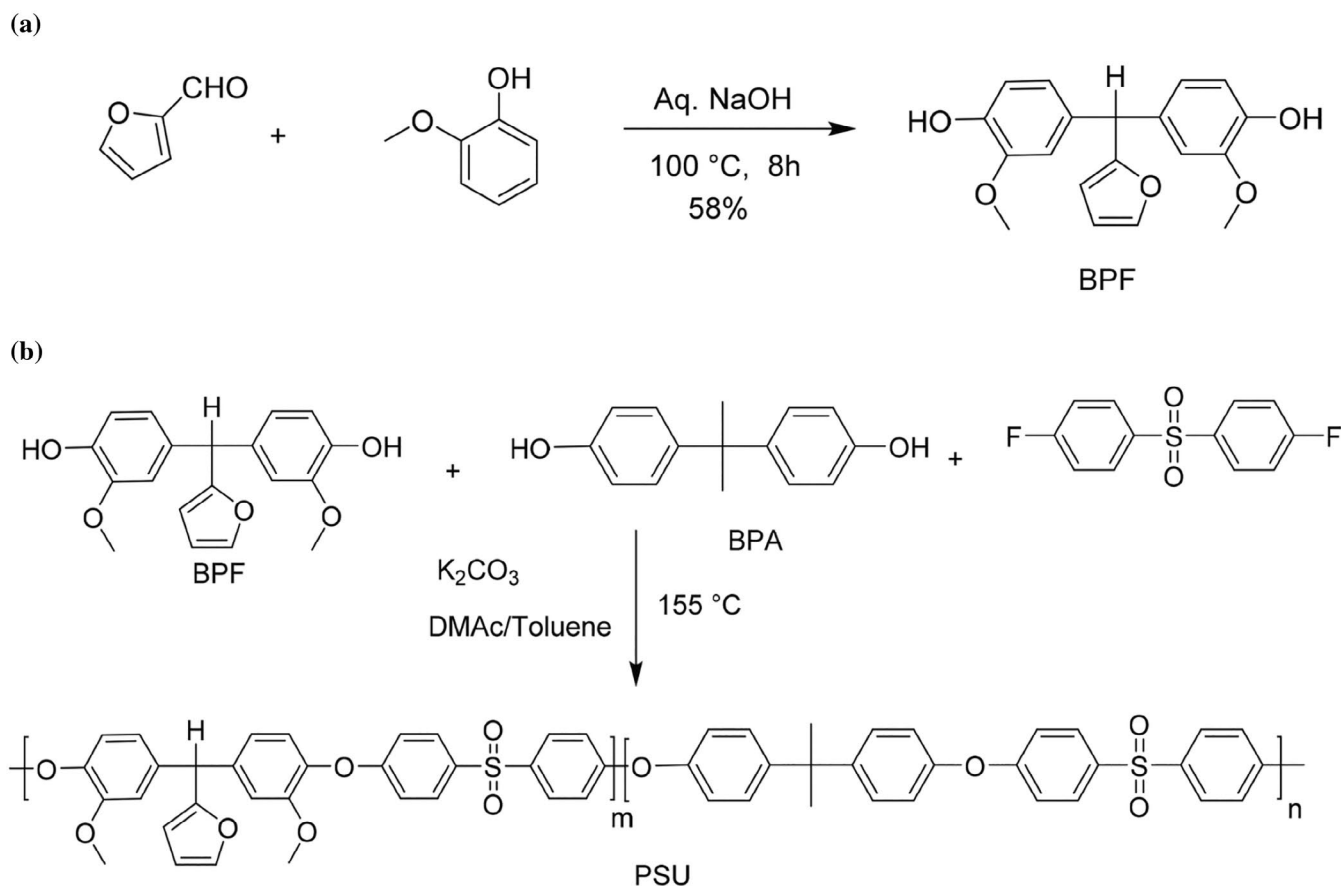


FIGURE 7 | (a) Synthesis of 4, 4'-(furan-2-ylmethylene)bis(2-methoxyphenol); (b) Synthesis of (co)polysulfones bearing pendant furyl groups [18].

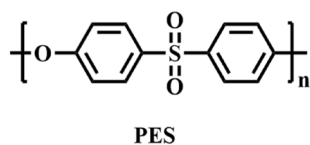


FIGURE 8 | Structure of PES.

high-molecular-weight polysulfones. As the focus on environmental sustainability grows, biomass-based green synthesis strategies will become a key research area in the future development of BPA-based polysulfones.

3 | Polyethersulfone (PES)

Polyethersulfone (PES) is an engineering thermoplastic characterized by alternating aromatic ether ($-\text{O}-$) and sulfone ($-\text{SO}_2-$) linkages within its polymer backbone. This high-performance material is synthesized via nucleophilic aromatic substitution polycondensation, typically employing 4,4'-dihydroxydiphenyl sulfone and dichlorodiphenyl ether as comonomers in the presence of alkaline catalysts in aprotic polar solvents. The ether linkages confer chain flexibility and melt-processability, while the rigid sulfone moieties impart exceptional thermal stability and mechanical robustness through resonance stabilization and dipolar interactions [23]. The chemical structure of polyethersulfone (Figure 8) is as follows:

Polyethersulfone (PES) demonstrates an elevated glass transition temperature (T_g) of 220°C–230°C, attributable to the restricted chain mobility imposed by its rigid-rod molecular architecture incorporating sulfone groups and aromatic rings. This thermal property profile enables sustained mechanical performance in elevated temperature environments exceeding 200°C. When compared to other polysulfone variants (e.g., PSF, PPSU), PES exhibits superior tensile strength (~85 MPa) and flexural modulus (~2.6 GPa), along with exceptional dimensional stability (coefficient of thermal expansion ~55 ppm/°C) due to its highly conjugated backbone structure [24].

PES materials are widely used in industries such as electronics, aerospace, and water treatment. For instance, PES is commonly employed in the production of high-performance filtration membranes [25], extensively used in ultrafiltration and reverse osmosis systems [26, 27] due to its outstanding chemical resistance and mechanical strength [28]. Furthermore, PES is used in the manufacturing of high-performance composites, insulating materials, and heat-resistant piping [29].

The preparation of polyethersulfone (PES) primarily employed traditional solution polymerization methods, which could be further classified based on different reaction mechanisms, such as nucleophilic aromatic substitution, self-condensation, and electrophilic substitution.

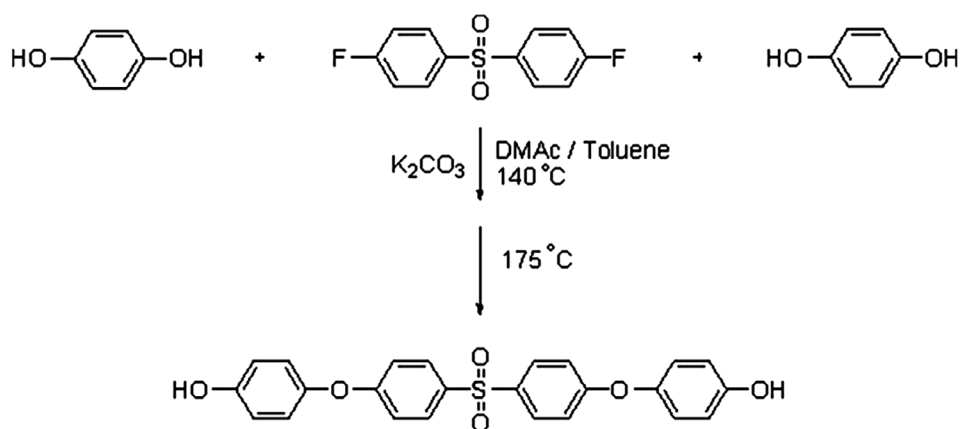


FIGURE 9 | Synthesis of 4,4'-(4,4'-sulfonylbis-(1,4-phenylene))bis(oxy) diphenol [30].

3.1 | Nucleophilic Aromatic Substitution (SNAr)

The most common method for the synthesis of polyethersulfone (PES) involves the reaction between a dihalogenated aromatic compound, such as dichlorodiphenyl ether, and a bisphenol, such as 4,4'-dihydroxydiphenyl sulfone, in the presence of a strong base. The SNAr reaction mechanism involves: (1) Deprotonation of bisphenolic hydroxyl groups to form phenolate nucleophiles. (2) Activation of the aromatic chloride moieties by the electron-withdrawing sulfone groups. (3) Nucleophilic attack at the activated aryl carbon (C—Cl bond) with concomitant chloride elimination. (4) Formation of stable ether linkages (—Ar—O—Ar—) through successive polycondensation steps.

Recent research presented a method for creating nanoporous PES membranes [30]. The precursor triblock copolymer, PLA-*b*-PES-*b*-PLA, was synthesized by first forming PES via SNAr, followed by ring-opening transesterification polymerization (ROTEP) of lactide to create the triblock structure. The sacrificial PLA block was then removed to generate nanoporous PES with slit-like pores. Characterization techniques such as small-angle X-ray scattering (SAXS) and transmission electron microscopy (TEM) confirmed the enhanced mechanical stability and ultrafiltration potential of the nanoporous membranes.

Further advancements in PES synthesis include the development of sulfonated poly(arylene ether sulfone) (SPES) membranes via SNAr polymerization [31]. By copolymerizing sulfonated and non-sulfonated monomers with aromatic diols (Figure 9), researchers achieved precise control over sulfonation degrees (40%, 50%, and 60%). The resulting membranes were cast and acid-treated to enhance proton conductivity, with higher degrees of sulfonation leading to increased proton conductivity and ion-exchange capacity. However, higher sulfonation levels also compromised thermal stability due to the degradation of the sulfonic acid groups, making these membranes particularly suitable for fuel cell applications. In addition, modern click chemistry technology also provides a multifunctional pathway for designing hydrophilicity and improving water-salt selectivity for PES membranes. Recently, Meng et al. demonstrated that thiol-ene click chemistry bridges various hydrophilic groups (carboxyl, cysteine, and amino groups) to allyl-containing polyethersulfone backbones. It is worth noting that the cysteine-modified variant (HPES-CYSAH) introduces oligomer properties,

showing excellent hydration ability, high free volume fraction, and excellent water-salt selectivity, which is superior to commercial sulfonated polysulfones [32].

In another study, fluorinated polyethersulfone (PES-6AF) copolymers were synthesized via SNAr reactions using 4,4'-dichlorophenyl sulfone (DCS), bisphenol S (BPS), and 6AF [33]. The incorporation of fluorine groups significantly lowered the dielectric constant while maintaining excellent thermal stability and mechanical strength. By varying the content of 6AF, the researchers were able to reduce the dielectric constant to as low as 2.5 at 1 kHz, rendering the copolymers suitable for high-frequency applications such as 5G systems and integrated circuits.

The development of hyperbranched polyethersulfone (HPES) from bisphenol S and 2,4,6-trifluorophenylsulfone via SNAr polymerization also offered promising advancements [34]. By controlling monomer ratios, both fluoro-terminated (F-HPES) and phenolic-terminated (OH-HPES) polymers could be synthesized. When blended with linear poly(ether sulfone) (LPES), HPES was found to reduce melt viscosity and improve high-temperature fluidity, enhancing the processing and coating properties of LPES without compromising its thermal or mechanical stability. This method held potential for improving the performance of LPES in industrial applications. In addition, the synthesis of highly branched poly(arylene ether sulfone) (HBPAES) via A2 + B3 polymerization and subsequent blending with linear poly(arylene ether sulfone) (LPAES) has gained attention [35]. By controlling the degree of branching in HBPAES, researchers explored the effects of branching on the morphology and mechanical properties of the blends. The results demonstrated that fine-tuning the branching structure of HBPAES can enhance mechanical performance and offer customizable properties for the blends, providing new opportunities for tailored material design.

3.2 | Self-Condensation

Polyethersulfone (PES) is a high-performance material renowned for its exceptional thermal stability, mechanical strength, and chemical resistance. Traditionally, PES was synthesized via polycondensation reactions. In recent years,

researchers have explored more efficient routes for PES synthesis through the self-polycondensation of AB-type monomers. This method offers not only high synthetic efficiency but also flexible molecular design options, making it an attractive alternative for large-scale industrial production.

The self-polycondensation of AB-type monomers provides an efficient route for synthesizing polyethersulfones (PES) through an intramolecular SNAr mechanism, where each monomer incorporates both nucleophilic and electrophilic functionalities within a single molecular architecture. This design ensures inherent 1:1 stoichiometric balance, enabling precise control over molecular weight development while eliminating the need for exogenous catalysts or coupling agents. The polymerization proceeds via the formation of a Meisenheimer complex intermediate, yielding either linear polymers with narrow polydispersity (PDI 1.2–1.5) when using symmetric AB monomers, or introducing controlled branching (10%–15%) through asymmetric monomer design [36].

The synthetic simplicity of this approach, characterized by high functional group conversion (>95%) and minimal byproduct formation, offers significant advantages for industrial-scale production. The absence of catalyst residues and reduced energy requirements (30%–40% lower than conventional methods) makes this method particularly suitable for manufacturing medical-grade PES with ultralow metal contamination (<5 ppm) and functionalized derivatives. Furthermore, the well-defined reaction kinetics and excellent scalability render this process adaptable to continuous production systems, meeting the growing demand for high-performance PES materials with tailored properties.

One study reported the successful synthesis of telechelic PES oligomers through chain-growth polycondensation [37]. By utilizing an AB monomer with 4-fluorophenyl-4'-methoxyphenyl sulfone as an initiator, oligomers with molecular weights ranging from 1600 to 2800 g/mol and low polydispersity were produced. This reaction followed a nucleophilic aromatic substitution mechanism, and the resulting oligomers were characterized using NMR and MALDI-TOF, demonstrating their potential for further polymerization. Another study focused on the self-polycondensation of AB₂-type macromonomers to create long-chain branched poly(ether sulfone) (PES) [38]. By introducing branching points within the polymer structure, the resulting PES exhibited higher molecular weights, improved chain entanglement, and enhanced mechanical and rheological properties than the linear PES, making it ideal for applications.

In a separate investigation, the synthesis of hyperbranched polyethersulfone via self-polycondensation was explored [39, 40]. Hyperbranched polymers exhibited a high degree of branching, which imparts excellent solubility, lower viscosity, and superior film-forming properties than linear polymers. The hyperbranched structure generated through self-polycondensation, also enhanced ion-exchange capacity and water flux in membranes, making these materials particularly promising for filtration and separation technologies (Figure 10).

Self-polycondensation of polyethersulfone offered greater design flexibility than traditional polycondensation methods. By

selecting the monomer structure, researchers could precisely control the molecular weight and architecture of the polymers. Additionally, this method generally required neither complex reaction conditions nor catalysts, contributing to its environmentally friendly nature. The intrinsic reactivity of AB-type monomers minimized by-product formation and improved reaction efficiency, making self-polycondensation a compelling method for developing high-performance PES materials.

3.3 | Aromatic Electrophilic Substitution Reaction

Another method for synthesizing PES is through an aromatic electrophilic substitution reaction, such as the Friedel-Crafts reaction. This process involves the polymerization of diphenyl sulfone with aromatic ether compounds under the catalysis of Lewis acids, such as AlCl₃ or BF₃. Although this method can produce PES, the reaction tends to generate branched structures, making it difficult to obtain highly linear polymers. This limitation reduces its viability for industrial applications.

Song and coworkers provided a detailed discussion on the application of Friedel-Crafts electrophilic substitution in the synthesis of polysulfones (Figure 11) [41]. The key to this reaction lay in the use of Lewis acids, such as aluminum chloride, as catalysts to facilitate the condensation between aromatic compounds and acyl halides or acyl chlorides, leading to the formation of high-performance polysulfone copolymers containing sulfone and ketone groups. For instance, 4,4'-diphenoxydiphenyl sulfone (DPODPS) and 4,4'-bis(4-phenoxybenzoyl) diphenyl (BPOBDP) underwent an electrophilic substitution reaction with isophthaloyl chloride (IPC) under the catalysis of anhydrous AlCl₃, resulting in the synthesis of polysulfone copolymers.

3.4 | Other Methods

Colquhoun and coworkers explored a novel method for the synthesis of aromatic ether-sulfones, using nickel-catalyzed coupling reactions [42]. This approach involved the coupling reaction of aryl dichlorides under pseudo-high-dilution conditions, forming highly strained macrocycles. These strained structures served as efficient precursors for ring-opening polymerization, allowing the production of high-molecular-weight polymers with superior thermal stability, mechanical strength, and impact resistance.

4 | Polyphenylsulfone (PPSU)

Polyphenylsulfone (PPSU) is a class of polymers synthesized through the polycondensation of aromatic diols and dichlorodiphenyl sulfone. The chemical structure of PPSU is as follows (Figure 12). Similar to PES, PPSU exhibits high thermal and chemical stability [43, 44], but with superior mechanical strength and wear resistance [45, 46]. In recent years, researchers have focused on modifying PPSU to enhance specific properties [47]. For example, sulfonation improves its ion-exchange capacity [48], making it more suitable for use in fuel cell membranes, while amination increases its hydrophilicity, leading

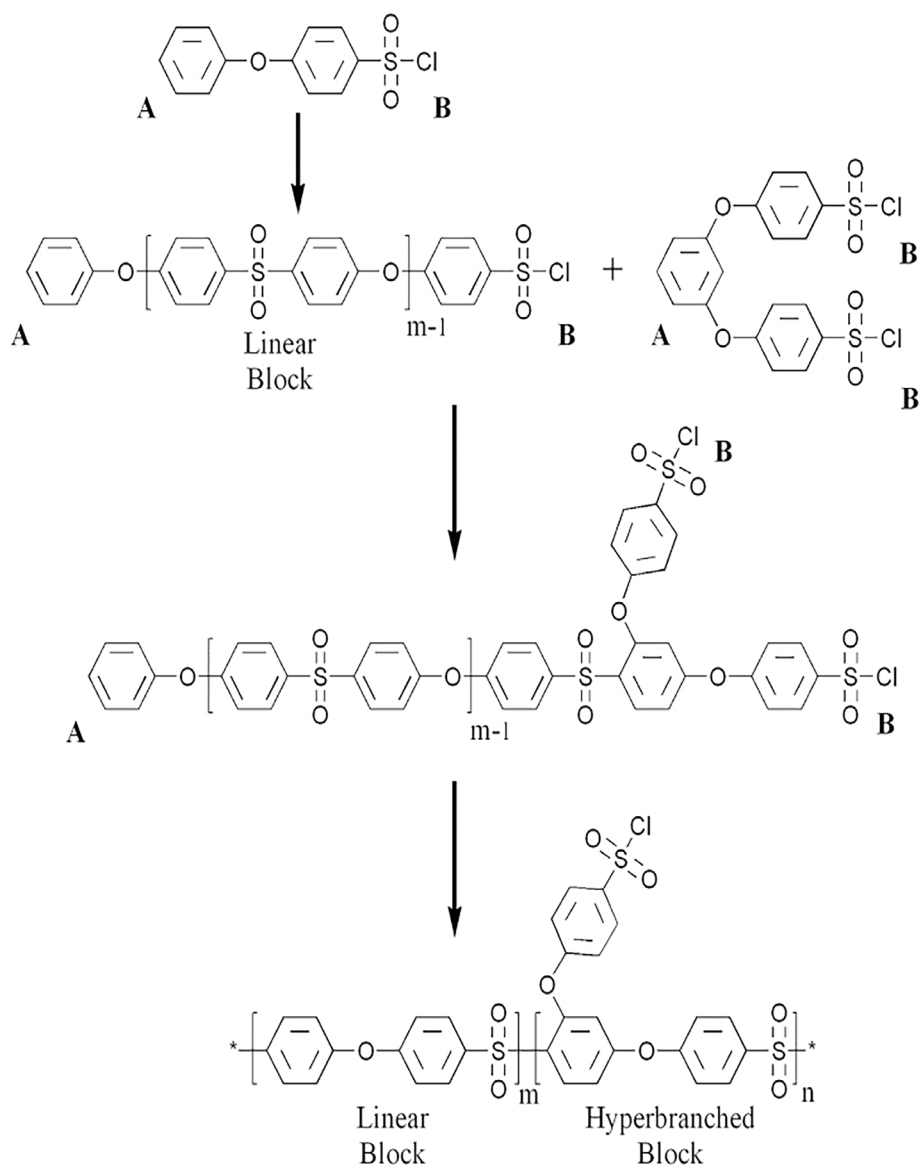


FIGURE 10 | Reaction scheme to generate linear-hyperbranched multiblock copolymers. “A” and “B” denote the electron-rich aromatic ring and electrophile, respectively. Dendritic and terminal repeat units of hyperbranched block are not shown [39].

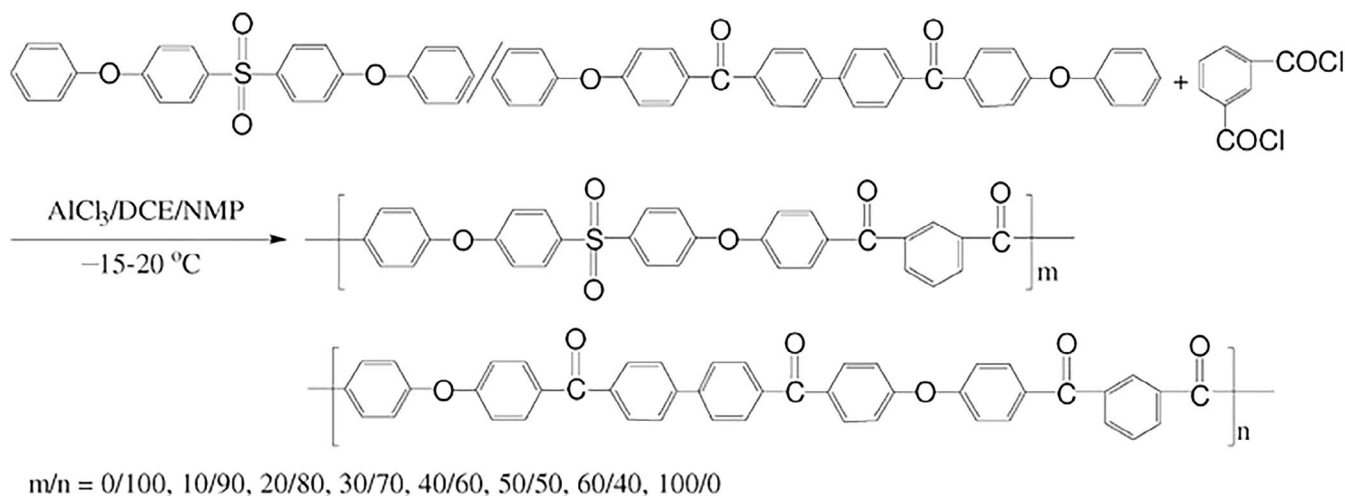


FIGURE 11 | Crafts electrophilic substitution in the synthesis of polysulfones [41].

to improved performance in water-treatment applications [49–51].

Polyphenylsulfone (PPSU) demonstrates outstanding thermal performance, maintaining dimensional stability and mechanical integrity at continuous service temperatures up to 200°C due to its rigid aromatic backbone and high T_g (~220°C). The polymer's unique macromolecular architecture, featuring alternating phenyl rings and sulfone groups, prevents melting behavior while preserving structural integrity under thermal stress. This thermal resilience, combined with exceptional hydrolysis resistance, enables PPSU to endure repeated steam sterilization cycles without significant property degradation, making it particularly valuable for medical applications including reusable surgical components and sterilization containment systems. The material's combination of high impact strength and optical clarity has led to its widespread adoption in demanding engineering applications. In the automotive sector, PPSU serves in under-the-hood components requiring both thermal stability and impact resistance, while in fluid handling systems, its chemical inertness and dimensional stability make it ideal for high-temperature piping applications [52]. These properties stem from the polymer's balanced molecular design, where the sulfone groups provide thermal and chemical resistance while the biphenyl units contribute to chain flexibility and optical properties.

4.1 | Nucleophilic Aromatic Substitution (SNAr)

Polyphenylsulfone (PPSU) is a high-performance polymer synthesized via aromatic nucleophilic substitution (SNAr), where electron-deficient aromatic halides, such as dichlorodiphenyl sulfone, react with aromatic diols. This process yields a polymer with excellent thermal and chemical stability, which is crucial for applications that demand resistance to harsh environments. Controlling the molecular weight of PPSU is essential for optimizing its mechanical and thermal properties, as demonstrated

by McGrath and his coworkers [53]. They synthesized disulfonated poly(arylene ether sulfone) copolymers with controlled molecular weights for fuel cell applications. By adjusting monomer ratios and reaction conditions, molecular weight distributions can be fine-tuned, directly affecting polymer performance.

Enhancing the chemical stability of PPSU, particularly for use in high-temperature environments, has been a major focus. He and coworkers demonstrated that the grafting of free radical scavengers onto poly(arylene ether sulfone) backbones could significantly improve the oxidative stability of polymer electrolyte membranes used in fuel cells [54]. This method showed the potential of chemical modifications to extend PPSU's durability in aggressive conditions. Similarly, Xu and coworkers explored the application of the crosslinked sulfonated PPSU membranes in direct methanol fuel cells (DMFCs) (Figure 13) [55]. Crosslinking not only enhances membrane durability but also reduces methanol crossover while maintaining high proton conductivity—a key feature for efficient fuel cell operation.

Another significant area of research involves the control of both branching and sulfonation in PPSU to improve its thermal stability and water retention properties. Ortenzi and coworkers developed a one-pot synthesis method that simultaneously adjusts the degree of branching and sulfonation in random sulfonated poly(arylene ether sulfone)s (Figure 14) [56]. This approach allowed for precise structural tuning, which enhances water retention while maintaining thermal stability—an important balance for proton exchange membrane (PEM) applications. Ionic crosslinking has also been explored as a way to improve PPSU's mechanical strength and chemical stability. Xu and coworkers developed ionically crosslinked sulfonated PPSU/polybenzimidazole (PBI) composite membranes using in situ polymerization [57]. This technique produced robust ionically crosslinked networks, improving membrane properties such as proton conductivity and thermal resistance, making these materials suitable for high-temperature PEM fuel cells.

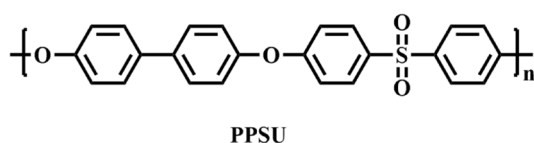


FIGURE 12 | Structure of PPSU.

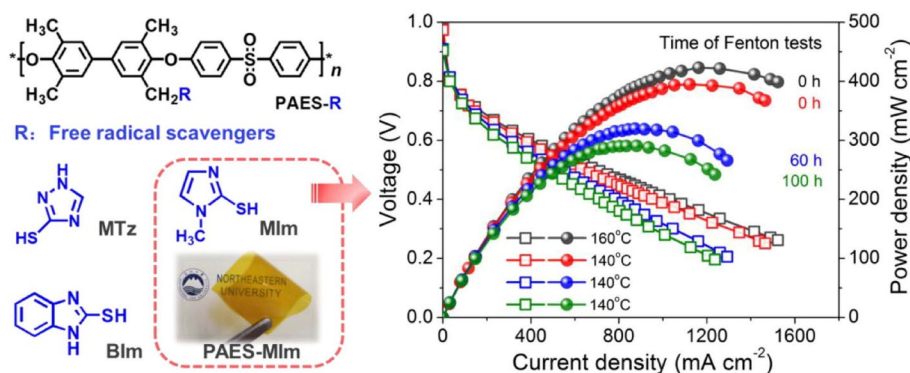


FIGURE 13 | Application of the crosslinked sulfonated PPSU membranes in direct methanol fuel cells [55].

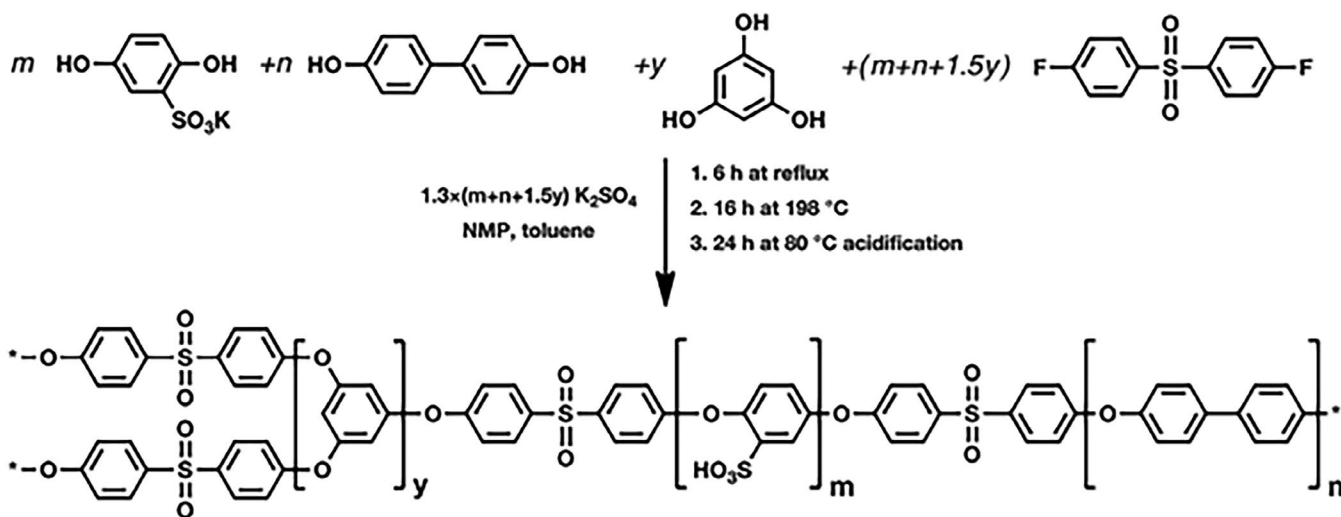


FIGURE 14 | The one-pot polymerization of branched sulfonated SPES [56].

coworkers synthesized sulfonated PPSU with a low degree of sulfonation, achieving membranes with high hydrophilicity while maintaining good structural integrity [58]. This lower degree of sulfonation enhances water interactions without sacrificing the material's mechanical properties.

Further advancements in PPSU modification have focused on the development of multiblock copolymers. McGrath and coworkers synthesized multiblock poly(arylene ether nitrile)-disulfonated poly(arylene ether sulfone) copolymers with enhanced proton conductivity and thermal stability [59]. Multiblock architectures enable more precise control of ion transport pathways, making these materials more efficient for proton exchange membranes. Similarly, Planes and coworkers developed partially fluorinated poly(arylene ether sulfone) multiblock copolymers with perfluorosulfonic groups, which improve chemical resistance and proton conductivity in high-performance PEMFCs [60]. The introduction of fluorine reduces water uptake and swelling, key factors for membranes used in high-temperature fuel cells.

Post-sulfonation also provides an effective method to control the properties of PPSU. Chen and coworkers synthesized multiblock sulfonated PPSU membranes using precise post-sulfonation, achieving materials with tailored thermal stability and proton conductivity [61]. This method offers flexibility in controlling the degree of sulfonation and the distribution of sulfonic acid groups, enabling the creation of PPSU membranes with highly specific properties for applications such as fuel cells and water purification.

In conclusion, aromatic nucleophilic substitution (S_NAr) remains a robust method for synthesizing high-performance PPSU with tunable properties. Through various modifications such as sulfonation, crosslinking, and the incorporation of multiblock structures, researchers have expanded the potential applications of PPSU in fields ranging from fuel cell membranes to water treatment. By controlling the molecular architecture, these advancements allow for fine-tuning the polymer's thermal, mechanical, and chemical

properties to meet the demands of increasingly specialized high-performance applications.

4.2 | Other Methods

Nickel-catalyzed coupling has emerged as an efficient method for synthesizing PPSU, offering enhanced control over polymer structure and properties, as compared to traditional nucleophilic aromatic substitution. This approach involved the coupling of aryl dichlorides, such as 4,4'-dichlorodiphenyl sulfone, in aprotic solvents like DMAc under inert conditions [62]. The use of zero-valent nickel catalysts with ligands such as triphenylphosphine ensured high molecular weight polymers with excellent thermal stability, toughness, and oxidative resistance.

5 | Conclusion

The evolution of PSU synthesis is driven by the need for efficiency, sustainability, and functionality. Solution polycondensation and melt polycondensation techniques have demonstrated versatility by allowing fine control over polymer molecular weight and architecture. Emerging methods such as vortex fluidic synthesis and phase transfer catalysis have offered new avenues for enhancing material properties while reducing environmental impact by eliminating toxic solvents.

In response to the growing emphasis on environmental sustainability, biomass-based green synthesis strategies have been explored, utilizing renewable resources such as lignin-based monomers to reduce dependency on petrochemical-derived materials. Chemical functionalization has further broadened the applications of polysulfones, particularly in membrane technologies and biomedical devices, by enhancing properties like hydrophilicity, biocompatibility, and ion-exchange capacity [63].

Additionally, advancements in the self-polycondensation of AB-type and AB₂-type monomers have enabled the synthesis of complex architectures, including hyperbranched and long-chain branched polymers. These developments offer enhanced mechanical and rheological properties, lower melt viscosity, and better film-forming capabilities, further expanding the potential of polysulfones in advanced technological applications.

In addition to the extensively reviewed membrane and biomedical fields, polysulfone has made significant advances in other high-performance areas. In the field of energy storage and conversion, sulfonated polysulfone is a key material for proton exchange membranes in fuel cells and vanadium redox flow batteries due to its tunable ionic conductivity and excellent chemical stability [9, 30, 53]. The high thermal stability and mechanical properties of polysulfone also make it a binder and diaphragm component for next-generation lithium batteries. The aerospace industry uses polysulfone for interior components, tubing, and as a substrate for advanced composites because of key advantages such as flame retardancy, high strength-to-weight ratio, and the ability to withstand continuous high-temperature operation [42]. In addition, polysulfones with ultra-low dielectric constants have been developed in response to the demand for printed circuit board substrates and integrated circuit packages for 5G communications and high-frequency electronics [11, 31]. Emerging applications also include additive manufacturing (3D printing) of durable, sterilizable parts, demonstrating the versatility of the material in an expanding industrial sector.

In summary, the field of polysulfone synthesis is experiencing rapid advancements with a focus on improving material performance and addressing environmental concerns. Traditional methods remain relevant, but novel approaches such as green synthesis and innovative polymerization techniques are becoming crucial for meeting industrial demands. The integration of sustainable practices, coupled with continued exploration of functional modifications, will shape the future development and application of polysulfones across diverse industries.

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Conflicts of Interest

The authors declare no conflicts of interest.

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